

Loop.exe – Game Design Document (Game Jam Edition)

Project Overview

Title: Loop.exe

Genre: Puzzle / Memory / Reaction

Engine: Unreal Engine 5

Platform: PC (Keyboard Only)

Development Context: GMTK Game Jam 2025

Role: Solo Developer

Development Time: ~4 Days (Submitted 1 day 6 hours before deadline)



Links :

Itch.io: <https://old-monkey.itch.io/loopexe>

YouTube: <https://www.youtube.com/watch?v=NZ2YteY25RY&t=1s>

Design Goal

A high-pressure memory execution game where difficulty is driven by **cognitive load and permanent consequences**, not mechanical complexity.

The player should feel like they are executing commands inside a failing system rather than playing a traditional puzzle game.

Design Pillars

1. **Cognitive Pressure Over Mechanics**

No movement, no camera control, no aiming. Difficulty comes purely from memory and precision.

2. **Permanent Consequences**

Mistakes delete execution threads permanently. There is no recovery during a run.

3. **Minimal Input, Maximum Tension**

Keyboard-only input with a single interaction stream.

4. **Retro-System Aesthetic**

CRT-inspired visuals and terminal-style presentation. The UI exists as part of the world.

Core Gameplay Loop

Generate Input Sequence

→ Display Sequence on CRT UI

→ Player Inputs Keys in Order

→ Correct Input → Sequence Extends

→ Incorrect Input → Thread Lost

→ All Threads Lost → Loop Purged (Game Over)

The loop continues indefinitely with increasing sequence complexity.

Player Interaction

Controls

- Keyboard only
- Enter to start the game
- Alphanumeric keys for sequence input

Player Constraints

- No pause
- No undo

- No retries within a run

These constraints enforce deliberate, focused play.

Thread System (Core Mechanic)

- Player starts with **3 execution threads**
- Each incorrect input removes **one thread permanently**
- Threads do not regenerate

Failure Condition

- All threads lost → immediate game over

Threads function as a representation of mental capacity rather than health.

Difficulty Scaling

Difficulty scales naturally through:

- Longer input sequences
- Increased cognitive load
- Player pressure due to accumulated progress

There are no artificial difficulty spikes or speed multipliers.

UI & Visual Design

CRT Simulation Approach

- Custom UI styled to mimic CRT glow, distortion, and scanlines
- UI rendered to a **material**
- Material applied to a **flattened curved cylinder mesh**
- This creates a physically curved screen effect instead of a flat overlay



The above image shows UI hiding inside tv and in left side **flattened curved cylinder mesh** with UI material applied on top of it

Documented Tradeoff

Rendering UI through a material introduces minor input latency, which slightly affects responsiveness during high-intensity sequences.

Future Improvement

- Replace material-based UI with flat UI + post-processing CRT effects
- Or hybrid approach using static CRT mesh and dynamic flat UI

Audio Design

Goals

- Reinforce tension
- Provide immediate feedback for correct and incorrect inputs
- Avoid cognitive overload

Sources

- Pixabay (royalty-free)

Hidden Content

- 10 hidden Easter eggs
- Optional discoveries for attentive players

- Do not affect the core gameplay loop
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Technical Constraints & Decisions

Constraints

- 4-day development window
- Solo development
- Unreal Engine 5 build pipeline

Key Decisions

- No save system
- No meta-progression
- No external services or plugins

Priority was given to stability and clarity of the core loop.

Build & Distribution Notes

- Final build size: ~470 MB
- PC downloadable build only

Identified Issue

Large file size reduced accessibility for jam players and discouraged quick play.

Future Approach

- Web-playable prototype
 - Aggressive asset trimming
 - Jam-first distribution mindset
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Postmortem Summary

What Worked

- Clear core mechanic
- Controlled scope
- Completed ahead of deadline

What Did Not Work

- Minor input latency due to UI material pipeline
- Oversized build for game jam context
- Familiar memory-game structure limited perceived creativity

Key Lesson

Accessibility and instant playability matter more than visual fidelity in game jams.

Credits

- Sound: Pixabay (royalty-free)
 - ASCII Art: <https://www.asciiart.eu/>
 - Environment Assets: Apartment Tech Props by Dekogon Studios
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Screenshot Placement Guide

Recommended screenshot locations:

1. After **Core Gameplay Loop** – gameplay in progress
2. Inside **UI & Visual Design** – CRT screen close-up
3. After **Thread System** – failure state / thread loss
4. Optional appendix at the end – full-screen visuals

Screenshots should support explanation, not decorate the document.